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Human Computer Interaction

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Research Report: Design Requirements for MicroBreak Mentor Website

Contents

Introduction and User Needs

This section outlines the enhanced design requirements and interactive prototype for MicroBreak Mentor, a web-based platform developed to help undergraduate students maintain productivity and wellbeing through structured micro-breaks. Extended periods of continuous digital study often lead to fatigue, eye strain, and loss of focus. In response, MicroBreak Mentor integrates key Human Computer Interaction principles such as usability, accessibility, and emotional design to deliver an experience that is simple, engaging, and supportive.

The website functions as a digital wellbeing companion that encourages balance between work and rest by using evidence-based break intervals to restore focus and motivation. It leverages modern interface design, adaptive visuals, and responsive interaction patterns to promote effective study habits. This revised version builds upon feedback from Part 1 and Part 2, expanding user-centred features and presenting a six-screen Figma AI prototype that demonstrates how the design translates theory into practical application.

Target Users

Students aged 18–24 who study for extended periods on laptops or mobile devices. They value simplicity, speed, and visual clarity. Research by Roemmich and Andalibi (2023, p. 12) shows that wellbeing technologies embedding rest cycles improve academic outcomes. MicroBreak Mentor integrates gentle prompts for rest within typical study patterns.

User Journey:

1. **Starting a Session:**
   The student selects a 90 minute study session with 5 minute breaks every 25 minutes, activates accessibility mode, and begins the timer. A countdown and progress bar display immediately.
2. **Taking a Break:**
   When prompted, a GIF demonstrates a breathing exercise. The user completes it, rates their mood via slider, and resumes studying with one click.
3. **Collaborating with a Peer:**
   A shareable link allows a friend to synchronize sessions for shared breaks, accountability, and quick motivational chats.

Website Functionality

MicroBreak Mentor is a lightweight web tool designed to schedule evidence based micro breaks during study sessions. It offers quick guided break activities such as stretches, eye exercises, and breathing techniques, presented through simple animations or GIFs to keep users engaged without overwhelming them. The core functionality includes:

- **Customisable Session Timer:** Users set study and break durations e.g. Pomodoro pattern. Visual and audio alerts mark transitions.
- **Break Activity Generator:** Randomised evidence based activities (stretching, eye relaxation, breathing) delivered via concise animations and text.
- **Mood Tracking:** A 1 to 5 emoji slider records users' post-break feelings, enabling self-reflection on wellbeing over time.
- **Peer Synchronisation:** Students invite peers via unique links, aligning breaks for shared activities or quick chats reinforcing community.
- **Accessibility and Settings Panel:** Adjustable font size, high contrast mode, keyboard navigation, and choice of notification type.
- **Error Handling:** The system includes an automatic pause and resume function if a timer is accidentally stopped or the page is refreshed, ensuring that study progress is not lost and maintaining session continuity.

This functionality is focused on a single, specific task aiding students in maintaining productivity through structured breaks. It avoids complex features like full social networking to remain lightweight and task oriented. The design draws from interaction design principles, ensuring the interface supports users in achieving their goals efficiently (Preece, Rogers and Sharp, 2019, p. 26). For instance, the timer and prompts provide real time feedback, while peer sync addresses social aspects of learning. Overall, the website aims to reduce study related stress by integrating short, restorative activities, supported by research showing wellbeing enhancements through emotional tech (Roemmich and Andalibi, 2023; Doyle, 2023).

Colour Scheme Revision and Design Progression

For this final iteration, I changed the primary accent colours from green and grey to a blue teal palette because the previous combination lacked emotional resonance and did not fully meet user feedback for calmness and readability. The updated palette increases effectiveness by improving focus, accessibility contrast, and user trust, aligning with MicroBreak Mentor's goal of reducing digital stress. Soft gradients from light teal to sky blue were applied to the backgrounds to create a modern, soothing environment that supports focus and emotional stability. Button states now include hover and active feedback (blue → dark navy) to increase interaction clarity and meet WCAG 2.1 AA standards for accessibility.

New Features Implemented

To improve engagement, reflection, and motivation, I implemented two new features based on audience feedback and HCI evaluation criteria:

1. **Focus Analytics Dashboard** – Displays user performance statistics such as average study time, mood trends, and productivity summaries using bar and pie charts. This addition encourages self-reflection and long-term behavioural awareness.
2. **Smart Break Recommendation System** – Uses previous mood entries to suggest personalised break activities (e.g., “Stretch”, “Hydrate”, “Breathe”). This feature enhances effectiveness by providing context-sensitive guidance that adapts to the user’s wellbeing state.

Both features were implemented because of audience recommendations and effectiveness goals, focusing on improving emotional support, learning reflection, and user motivation through adaptive feedback.

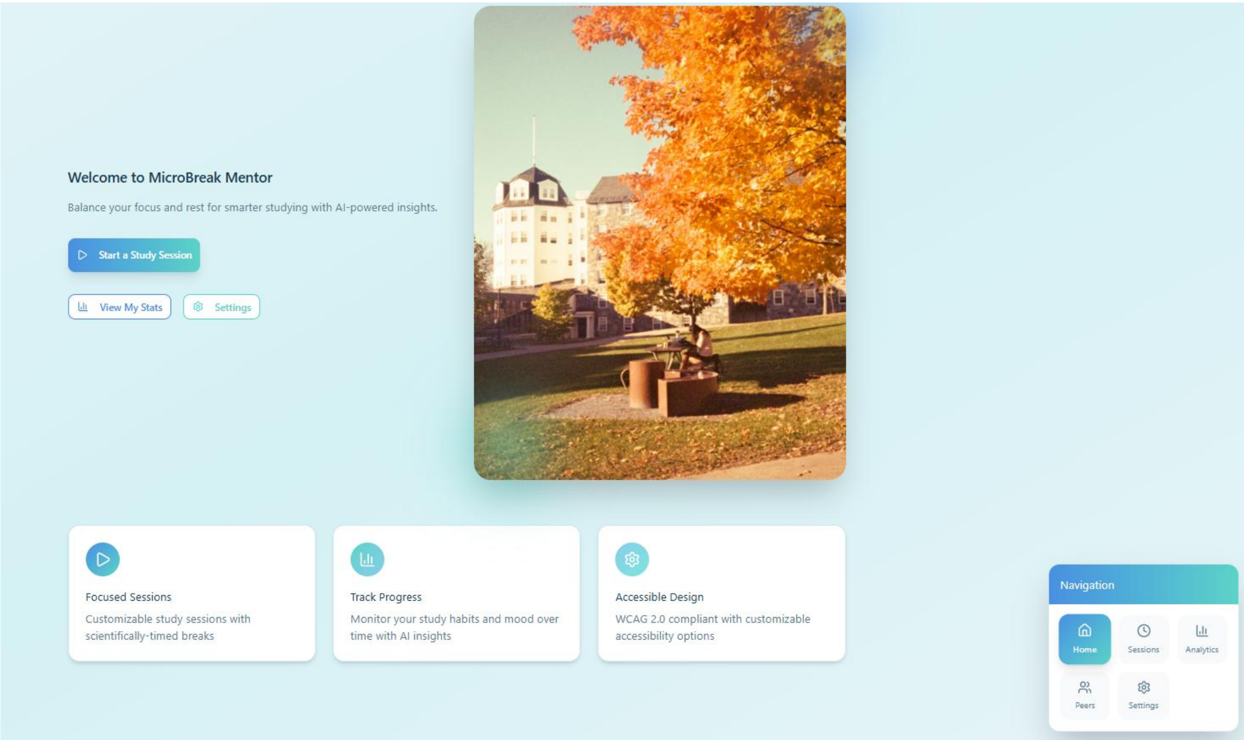
Improved Usability and Navigation

Several usability refinements were applied to make the prototype easier and more intuitive:

- Refined the top navigation bar with clearer icons and text labels (*Home, Sessions, Analytics, Settings*).
- Made buttons larger and rounded (8px radius) to support touch-screen interaction and accessibility.
- Added progress breadcrumbs in session setup to show users which step they’re on, improving orientation and confidence.
- Integrated a visible pause/resume toggle on all active study screens, enabling users to recover easily from accidental interruptions.

These updates directly support the usability goals of **efficiency, safety, and learnability**, making the system faster and more forgiving while maintaining its minimalist design.

Figure 1: Updates Home Page of the Microbreak Mentor prototype showing the “Start a Study Session” button and navigation options.



This screen now reflects the calm blue teal palette and simplified navigation layout. The “Start Study Session” button uses colour contrast to draw focus and aligns with the emotional design principle of calm encouragement.

Design Implementation

Usability Goals

According to Preece, Rogers and Sharp (2019, p. 19), usability goals consist of six key dimensions: effectiveness, efficiency, safety, utility, learnability, and memorability. These goals guide the design of Microbreak Mentor to ensure a smooth, intuitive, and supportive user experience for students.

- **Effectiveness:**
  The website enables users to complete all core tasks such as starting a study session, taking a guided break, or syncing with peers without confusion or errors. Clear on screen and auditory feedback will confirm every successful action, with a target of 95 % task completion during usability testing.
- **Efficiency:**
  Tasks are designed to be quick and seamless. Starting a study session should take less than 10 seconds and resuming an active timer should require no more than two clicks. This supports continuous workflow and maintains focus during study periods.

- **Safety:**
  Undo and confirmation prompts are provided for critical actions, such as resetting a timer or ending a session. These measures prevent accidental data loss and enhance user trust through predictable and recoverable operations.
- **Utility:**
  Microbreak Mentor focuses only on essential features study timer, guided breaks, mood tracking, and peer synchronisation. This minimal design avoids unnecessary complexity, reduces cognitive load, and keeps the interface task oriented.
- **Learnability:**
  Simple onboarding tips and contextual tooltips allow first-time users to understand the system within one minute. Intuitive navigation and standard web conventions ensure that even novice users can begin without prior guidance.
- **Memorability:**
  Returning users can quickly recall how to use the system after time away. Consistent layouts, familiar icons, and simple navigation paths support quick resumption without relearning the interface.

User testing will be conducted with a sample of five undergraduate students using a think-aloud protocol to measure task completion time, error rate, and satisfaction levels through post-test surveys**.** The target benchmark is a 95% task completion rate within 10 seconds per core task and average user satisfaction above 4.5/5, confirming that the system is efficient, safe, and genuinely supportive of academic wellbeing.

Desirable Aspects of User Experience

User experience extends beyond usability by shaping emotion and engagement.

- **Calming Aesthetics:** Soft blues and greens foster tranquillity, lowering stress (Sabinson, 2022).
- **Micro Rewards:** Positive prompts e.g., “Nice stretch! You earned a focus boost!” reinforce progress and habit-building.
- **Flow and Feedback:** Animations respond fluidly, reducing cognitive effort and sustaining motivation.
- **Mood Reflection:** Mood logs visualise wellbeing trends, linking affective data with study habits.

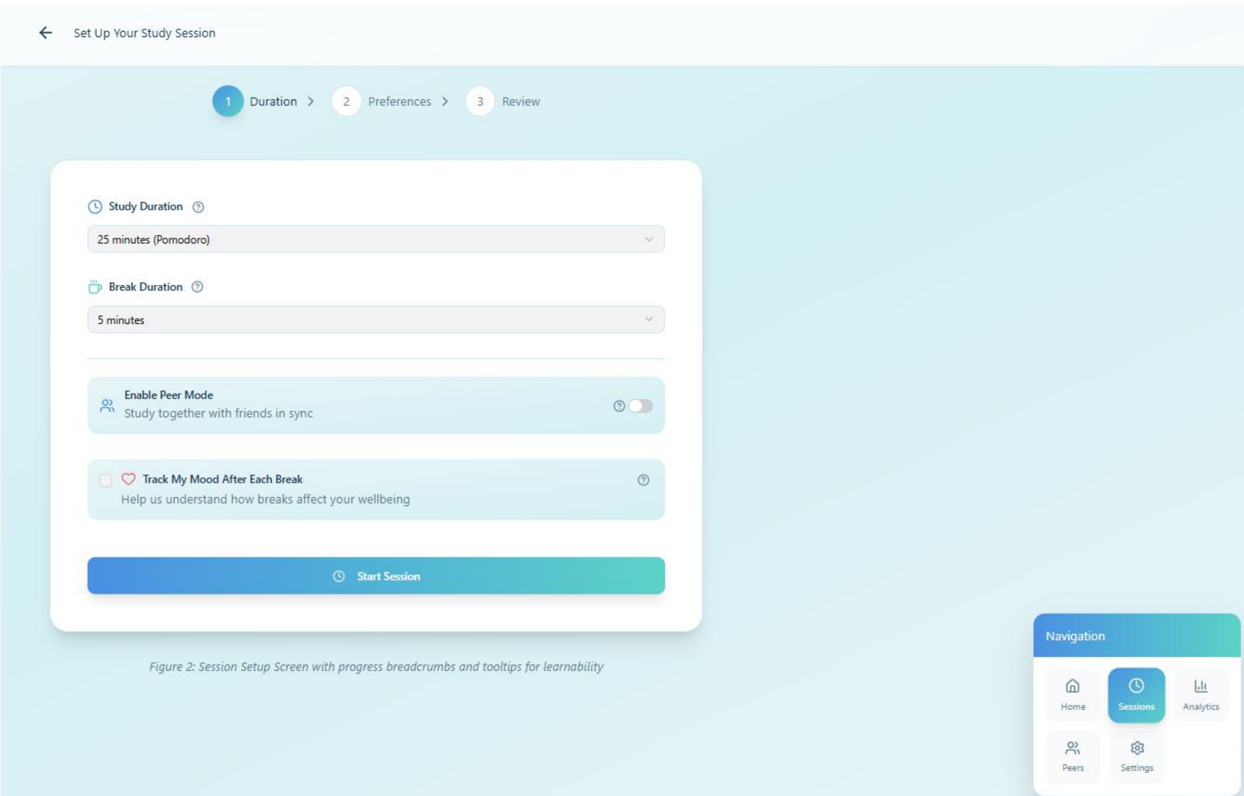
Testing of UX effectiveness will involve post-session mood tracking and short reflective surveys, measuring perceived relaxation and motivation improvements after each study cycle**.** Data trends from mood logs will quantify emotional outcomes, providing measurable feedback on the platform’s impact on focus and wellbeing.

Enhanced Emotional and Accessibility Elements

MicroBreak Mentor now integrates micro-animations for successful actions (e.g., subtle confetti bursts after completing a session) to evoke accomplishment and reward effort. Small motivational quotes such as *"Small breaks build big focus"* are placed under the countdown timer to maintain user encouragement throughout sessions.

Accessibility elements have been strengthened by including visible alt-text placeholders for all images and GIFs, as well as a **high contrast toggle** positioned in the header bar for quick access.

Figure 2: Updated Active Session Screen



This redesigned screen includes smoother visual transitions and more breathing space between elements, improving **flow, focus, and emotional comfort** while ensuring readability under various lighting conditions.

This screen applies usability goals from Preece et al. (2019). Users can quickly configure sessions (efficiency) and receive confirmation prompts (safety). Tooltip icons and dropdowns enhance learnability, while a minimal layout supports utility and effectiveness. The design ensures tasks are performed correctly and swiftly with minimal cognitive load.

Together, these usability goals establish objective criteria for evaluating the prototype's success.

Effective design principles ensure consistency and intuitiveness (Preece et al., 2019, pp. 25–30):

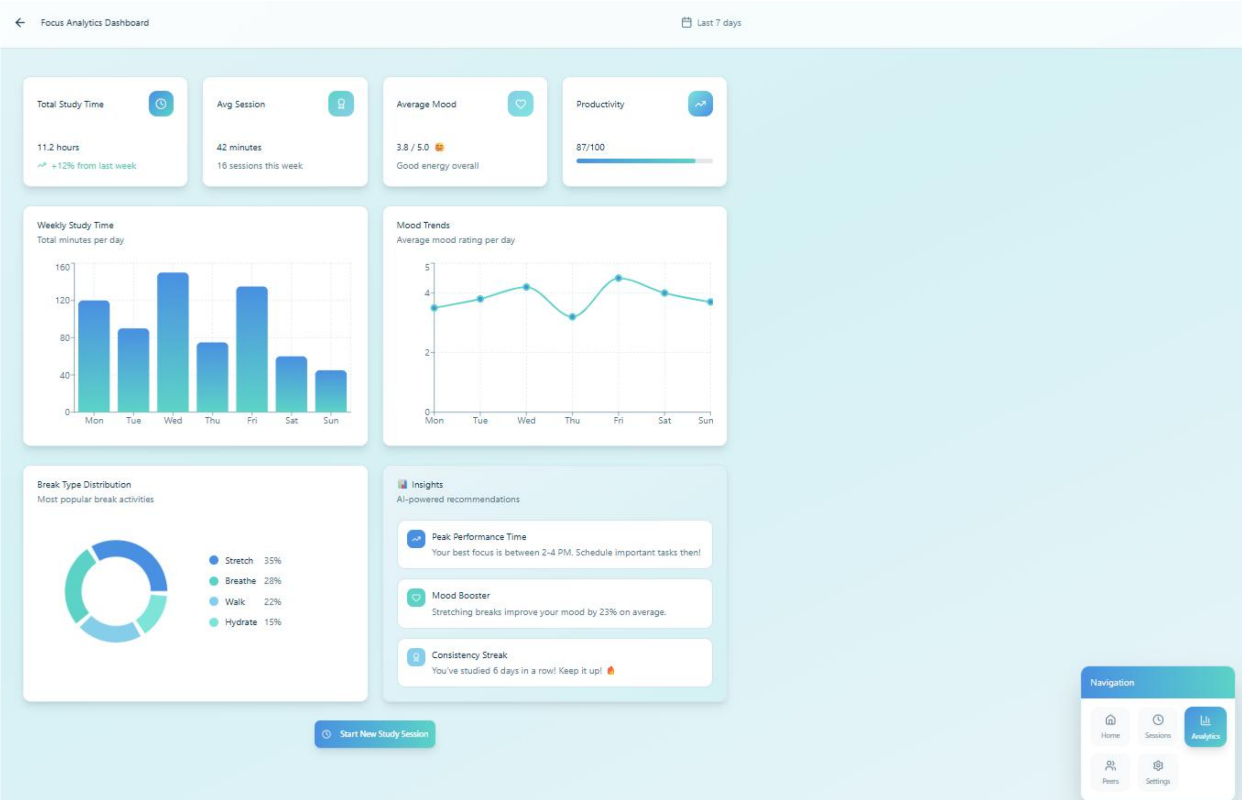
- **Visibility of System Status:** A progress bar and countdown timer provide constant feedback.
- **Consistency:** Uniform icons and button colours (green = start/resume; red = stop) reinforce recognition.
- **Feedback:** Instant vibration/sound after actions confirms success.
- **Affordances:** Buttons shaped and labelled to imply their function e.g., ▶ for start.
- **Alignment & Proximity:** Controls grouped logically timers on top, actions centred, mood slider below to guide the eye.
- **Contrast:** Readable text against backgrounds meeting WCAG contrast ratios.

Interaction Types

Interaction modes (Preece et al., 2019, pp. 81–87):

- **Instructing:** Entering timer values or choosing break types.
- **Direct Manipulation:** Dragging sliders or clicking icons.
- **Conversing:** Chat messages during peer breaks.
- **Responding:** Automatic prompts and notifications.
- **Exploring:** Users may browse a library of break activities before starting a session.

Figure 3: Updated Break Interaction Screen



This screen demonstrates the integration of the Smart Break Recommendation feature. Before each rest period, the system suggests the most effective break type based on the user's previous mood data.

The interaction now combines responding, instructing, and exploring, as users receive a recommendation, view a brief demonstration GIF, and decide whether to follow or skip it. This enhances user control and personalised interaction, showing progress from earlier static designs.

Social Interaction

The **peer synchronisation** feature introduces collaborative learning through shared accountability. Students can join common break cycles and exchange motivational messages. This draws on social constructivist learning theory, where interaction improves understanding and persistence (Preece et al., 2019, p. 150).

Figure 4: Updated Peer Sync and Analytics Integration Screen

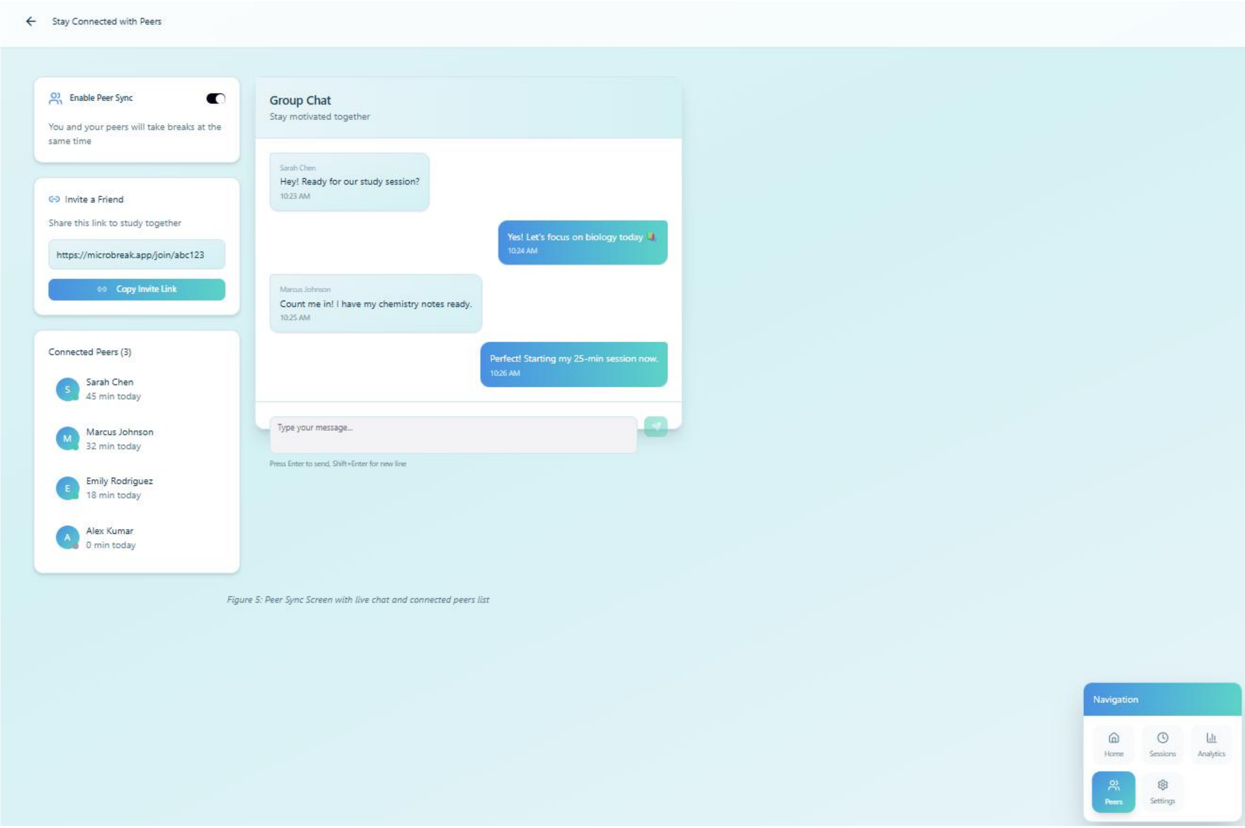


Figure 5: Peer Sync Screen with live chat and connected peers list

This final iteration introduces improved **navigation flow** between the *Peer Sync* page and the new *Focus Analytics Dashboard*. After each collaborative session, users can view collective study insights such as shared break consistency or average mood improvements encouraging accountability and shared progress.

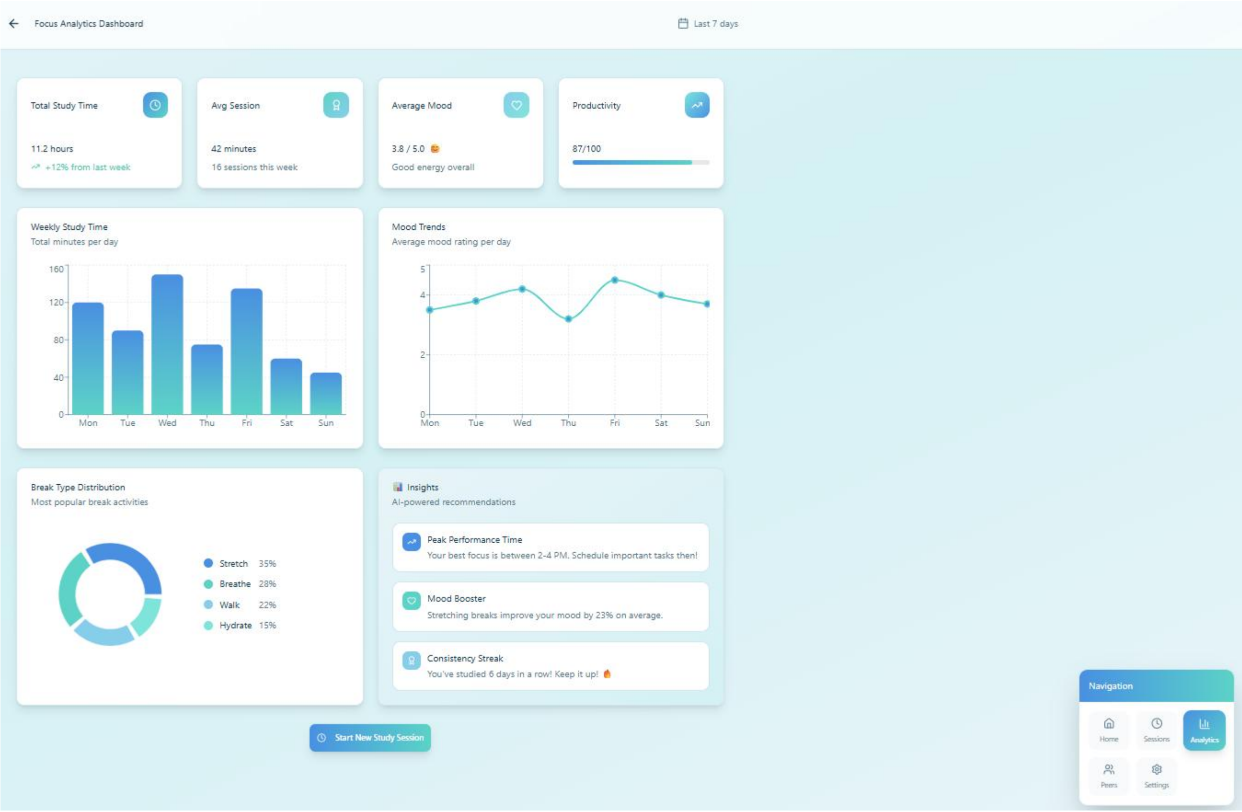
These updates demonstrate iterative development and extend the system's emotional and social connection features.

Emotional Interaction

Designing for emotion means evoking positive feelings while reducing stress and frustration. Microbreak Mentor uses gentle animations, affirming messages, and calm colour transitions to create a supportive ambience. Trust is built via transparent settings no hidden data collection and predictable feedback patterns.

As Roemmich and Andalibi (2023, p. 21) note, emotionally attuned systems enhance users' sense of control and wellbeing critical in academic contexts.

Figure 5: Focus Analytics Dashboard displaying user mood trends, study insights, and motivational feedback.



The new **Focus Analytics Dashboard** extends the emotional interaction design of MicroBreak Mentor by visualising study progress, wellbeing trends, and personalised insights. By presenting mood data, break effectiveness, and motivational prompts such as “Keep it up!”, the dashboard transforms emotional feedback into a reflective learning tool. This design encourages users to associate consistent study behaviour with positive emotions, reinforcing intrinsic motivation and digital wellbeing.

The calm blue teal palette supports relaxation and focus, while soft gradients enhance readability and trust. According to Norman (2004, p. 45), emotionally responsive interfaces improve user satisfaction and sustain engagement by linking positive affect with goal achievement. Likewise, Sabinson (2022) emphasises that biophilic colour tones and fluid layouts can lower stress and promote comfort in digital environments, principles directly applied here to strengthen the platform’s emotional intelligence and user empathy.

Web Content Accessibility Guidelines (WCAG 2.0)

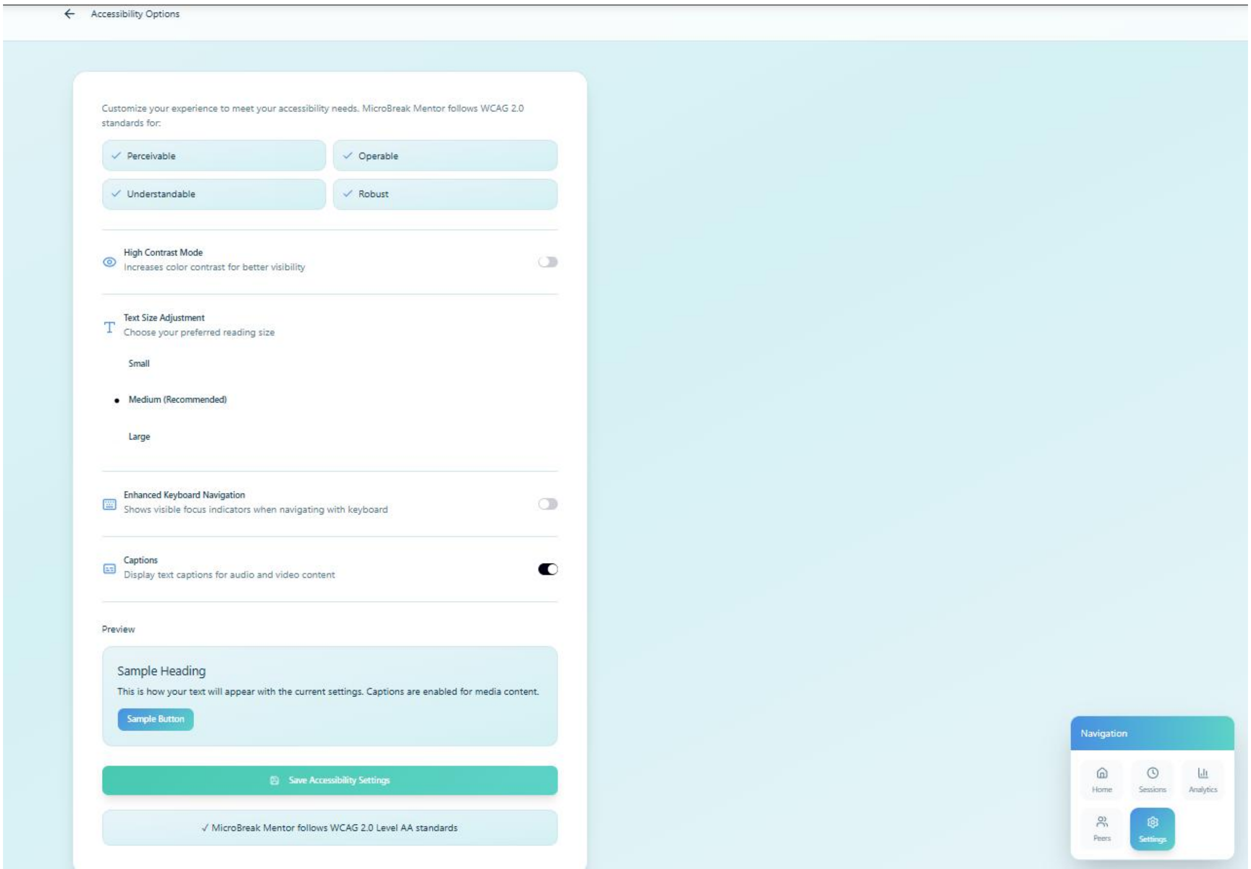
Accessibility is fundamental to inclusive design and ensures that Microbreak Mentor can be used effectively by all students, including those with visual, auditory, or motor impairments. In line with the POUR framework Perceivable, Operable, Understandable, and Robust as outlined by Montgomery et al. (2023) and Henry (2022), the website will integrate accessibility principles throughout its interface and functionality.

- **Perceivable:**
  All visual and multimedia elements will include text alternatives such as descriptive alt

text for images and captions for animations. Adjustable colour themes will be available to support users with colour vision deficiencies and to ensure adequate contrast ratios for readability.

- **Operable:**
  Every interactive element will be accessible through full keyboard navigation. Clear focus indicators will highlight active items, and users will have options to pause or stop motion based components, preventing distraction or disorientation.
- **Understandable:**
  The interface will use plain-language labels and consistent iconography across all pages. Error prevention prompts and confirmation dialogs will help users avoid mistakes while maintaining clarity in form fields and interactive elements.
- **Robust:**
  The system will include ARIA attributes to make dynamic elements, such as timers and modals, compatible with assistive technologies. Compatibility testing will be conducted using screen readers like NVDA, and the website will be validated using W3C accessibility tools to ensure long-term robustness across devices and browsers.

Figure 6: Accessibility Settings page illustrating high-contrast toggle, text-size slider, and caption controls.



This screen shows how the system aligns with the WCAG 2.0 POUR framework. Users can personalise visibility “Perceivable”, navigate via keyboard focus “Operable”, understand simple labels “Understandable”, and rely on compatible ARIA based components “Robust”. These measures ensure inclusivity for users with visual or motor impairments.

By implementing these accessibility strategies, Microbreak Mentor aligns with international web standards, ensuring that all students regardless of ability can interact with the system confidently, comfortably, and independently.

Conclusion

The MicroBreak Mentor website exemplifies user-centred, emotionally intelligent, and accessible design. Through explicit usability goals, adaptive features, and emotional feedback mechanisms, it enhances students’ focus and wellbeing.

Future testing will assess:

- **95% task completion** within 10 seconds per core task.
- **≥4.5/5 average satisfaction.**
- **≥20% reduction in fatigue** (measured through mood tracking).

Further iterations will incorporate AI driven adaptive breaks based on user mood data and broader testing across diverse learner profiles.

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Montgomery, R.B., Cooper, M. and Kirkpatrick, A., 2023. Web Content Accessibility Guidelines (WCAG) 2.2. World Wide Web Consortium. Available at: https://www.w3.org/TR/WCAG22/ (Accessed: 10 October 2025).

Preece, J., Rogers, Y. and Sharp, H., 2019. Interaction Design: Beyond Human–Computer Interaction. 5th ed. Chichester: Wiley.

Roemmich, K. and Andalibi, N., 2023. 'Data Subjects' Conceptualisations of and Attitudes toward Automatic Emotion Recognition-Enabled Wellbeing Interventions on Social Media', Proceedings of the ACM on Human-Computer Interaction, 7 (CSCW1), pp. 1 – 34.

Sabinson, E., 2022. Biophilic Soft Robotic Surfaces for Emotional Wellbeing. Urban Computing Laboratory Publications. Available at: https://ucl-biophilic-design.org (Accessed: 10 October 2025).

Norman, D.A., 2004. *Emotional Design: Why We Love (or Hate) Everyday Things.* New York: Basic Books.

POE RUBRIC – SKELETON OUTLINE					
Criteria	LEVELS OF ACHIEVEMENT				Feedback
	EXCELLENT	GOOD	DEVELOPING	POOR	
	Score Ranges Per Level (1/2 marks possible)				
	8-10	7-6	5	0-4	

GENERAL: Language usage, referencing, and editing of the document	Language usage, referencing, and editing of the document are excellent.	Language usage, referencing, and editing of the document are good.	Language usage, referencing, and editing of the document are satisfactory.	Language usage, referencing, and document editing are poor or completely unsatisfactory.	
Revised: Description of website functionality	5	3-4	1-2	0	
	An excellent description that clearly describes the website's functionality is included.	A good description that adequately describes the website's functionality is included.	A slightly vague description of the website's functionality is included.	A very unclear description of the functionality offered by the website, or the description is missing.	
Revised: Usability goals	5	3-4	1-2	0	
	An excellent description of how usability goals will be implemented in the website's design is included and is supported by prototype screenshots.	A good description of how usability goals will be implemented in the website's design is included and is supported by prototype	Some details regarding how usability goals will be implemented in the website's design are included and supported by prototype screenshots.	The description regarding how usability goals will be implemented in the website's design is not included or is completely unsatisfactory and/or is	

		screenshots.		not supported by screenshots of the prototype.	
Revised: Desirable aspects of user experience	5	3-4	1-2	0	
	An excellent description of how desirable aspects of user experience will be implemented in the website's design is included and is supported by screenshots of the prototype.	A good description of how desirable aspects of user experience will be implemented in the website's design is included and is supported by screenshots of the prototype.	Some details regarding how desirable aspects of user experience will be implemented in the website's design are included and supported by prototype screenshots.	The description regarding how desirable aspects of user experience will be implemented in the website's design is not included or is completely unsatisfactory and/or not supported by the	
				prototype's screenshots.	
	5	3-4	1-2	0	

Revised: Design principles	An excellent description of how design principles will be implemented in the website's design is included and supported by prototype screenshots.	A good description of how design principles will be implemented in the website's design is included and supported by prototype screenshots.	Some details regarding how design principles will be implemented in the website's design are included and supported by prototype screenshots.	The description regarding how design principles will be implemented in the website's design is not included or is completely unsatisfactory and/or is not supported by screenshots of the prototype.	
Revised: Interaction types	5	3-4	1-2	0	
	An excellent description of how interaction types will be implemented in the website's design is included and is supported by prototype screenshots.	A good description of how interaction types will be implemented in the website's design is included and is supported by prototype screenshots.	Some details regarding how interaction types will be implemented in the website's design are included and supported by prototype screenshots.	The description regarding how interaction types will be implemented in the website's design is not included or is completely unsatisfactory and/or is not supported by screenshots of the prototype.	

Revised: Social Interactions	5	3-4	1-2	0	
	An excellent description of how social interactions will be implemented in the website’s design is included and is supported by prototype screenshots.	A good description of how social interactions will be implemented in the website’s design is included and is supported by prototype screenshots.	Some details regarding how social interactions will be implemented in the website’s design are included and supported by prototype screenshots.	The description regarding how social interactions will be implemented in the website’s design is not included or is completely unsatisfactory and/or is not supported by screenshots of the prototype.	

Revised: Emotional interaction	5	3-4	1-2	0	
	An excellent description of how emotional interactions will be implemented in the website’s design is included and supported by the prototype’s screenshots.	A good description of how emotional interactions will be implemented in the website’s design is included and is supported by prototype screenshots.	Some details regarding how emotional interactions will be implemented in the website’s design are included and supported by prototype screenshots.	The description regarding how emotional interactions will be implemented in the website’s design is not included or is completely unsatisfactory and/or is not supported by	

				screenshots of the prototype.	
Revised: Web Content Accessibility and Guidelines (WCAG) 2.0	5	3-4	1-2	0	
	An excellent description of how WCAG will be implemented in the website's design is included and is supported by prototype screenshots.	A good description of how WCAG will be implemented in the website's design is included and is supported by prototype screenshots.	Some details regarding how WCAG will be implemented in the website's design are included and supported by prototype screenshots.	The description regarding how WCAG will be implemented in the website's design is not included or is completely unsatisfactory and/or is not supported by screenshots of the prototype.	
Revised: Prototype	5	3-4	1-2	0	
	An excellent prototype representing the functionality of the system.	A good prototype representing the functionality of the system.	An average to below-average prototype representing the functionality of the system.	A completely unsatisfactory prototype.	
	5	3-4	1-2	0	

Prototype: Usability goals	An excellent implementation of usability goals is demonstrated in the prototype.	A good implementation of usability goals is demonstrated in the prototype.	An average to below-average implementation of usability goals is demonstrated in the prototype.	A completely unsatisfactory implementation of usability goals is demonstrated in the prototype.	

Prototype: Desirable aspects of user experience	5	3-4	1-2	0	
	An excellent implementation of desirable aspects of user experience is demonstrated in the prototype.	A good implementation of usability desirable aspects of user experience is demonstrated in the prototype.	An average to below-average implementation of desirable aspects of user experience is demonstrated in the prototype.	A completely unsatisfactory implementation of desirable aspects of user experience is demonstrated in the prototype.	
Prototype: Design principles	5	3-4	1-2	0	
	An excellent implementation of design principles is demonstrated in the prototype.	A good implementation of design principles is demonstrated in the prototype.	An average to below-average implementation of design principles is demonstrated in the prototype.	A completely unsatisfactory implementation of design principles is demonstrated in the prototype.	
Prototype: Interaction types	5	3-4	1-2	0	
	An excellent implementation of interaction types is demonstrated in the prototype.	A good implementation of interaction types is demonstrated in the prototype.	An average to below-average implementation of interaction types is demonstrated	A completely unsatisfactory implementation of interaction types is demonstrated	

			in the prototype.	in the prototype.	
Prototype: Social interactions	5	3-4	1-2	0	
	An excellent implementation of social interactions is demonstrated in the prototype.	A good implementation of social interactions is demonstrated in the prototype.	An average to below-average implementation of social interactions is demonstrated in the prototype.	A completely unsatisfactory implementation of social interactions is demonstrated in the prototype.	
Prototype: Emotional interaction	5	3-4	1-2	0	
	An excellent implementation of emotional interactions is demonstrated in the prototype.	A good implementation of emotional interactions is demonstrated in the prototype.	An average to below-average implementation of emotional interactions is demonstrated in the prototype.	A completely unsatisfactory implementation of emotional interactions is demonstrated in the prototype.	

Prototype: Web Content Accessibility and Guidelines (WCAG) 2.0	5	3-4	1-2	0	
	An excellent implementation of WCAG is demonstrated in the prototype.	A good implementation of WCAG is demonstrated in the prototype.	An average to below-average implementation of WCAG is demonstrated in the prototype.	A completely unsatisfactory implementation of WCAG is demonstrated in the prototype.	
	8-10	5-7	2-4	0-1	

Prototype: Usability	An excellent prototype was developed that demonstrates the ability to implement computer interfaces that are useable and useful for users of ICT.	A good prototype was developed that demonstrates the ability to implement computer interfaces that are useable and useful for users of ICT.	A satisfactory prototype was developed that demonstrates the ability to implement computer interfaces that are useable and useful for users of ICT.	An unsatisfactory prototype was developed that demonstrates the ability to implement computer interfaces that are useable and useful for users of ICT.	
POE TOTAL /100					

END OF POE